Evaluation Report & Next-Step Execution Roadmap

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Assigned Construct: Processing Speed (Gs)

Proposed Task: Advanced Perceptual Speed Matrix

Evaluation Status: Proceed with Modifications and Controlled Complexity

1. Executive Summary

The proposed Advanced Perceptual Speed Matrix demonstrates excellent alignment with our vision of building an AI-driven cognitive assessment platform that measures cognitive efficiency through interactive and adaptive experiences. The task requires students to rapidly identify identical symbols and alphanumeric patterns under time constraints, making it engaging, scalable, and suitable for generating rich behavioral analytics. However, if the stimuli become excessively long or complex, the task may begin measuring Sustained Attention and Working Memory rather than Processing Speed. To align with our startup objectives, the assessment should prioritize rapid visual scanning, perceptual discrimination, and speed of classification while minimizing memory load and unnecessary reasoning demands.

2. Alignment with MentiScope's Objectives

• Alignment with CHC-Gs Theory: 9.5/10

• Scientific Validity: 9/10

• Innovation: 9/10

• Adaptive Potential: 9.5/10

• AI Readiness: 9.5/10

• MVP Feasibility: 9.5/10

• Overall Evaluation: 9.3/10 – Proceed with Modifications

3. Scientific Assessment

Primary Processing Speed components to measure:
• Perceptual Speed
• Visual Scanning Efficiency
• Rapid Classification
• Symbol Comparison Speed
• Speed–Accuracy Trade-off

Potential construct contamination:
• Sustained Attention

• Working Memory (Gsm)
• Visual Complexity Effects
• Strategy and Search Behaviours

The task should primarily assess how quickly students perceive, discriminate, and classify visual information while minimizing memory demands.

4. Recommended Task Architecture

Module 1 – Symbol Matching
Students rapidly identify identical symbols among simple distractors.
Measures: Perceptual speed and visual scanning efficiency.

Module 2 – Alphanumeric Comparison
Students identify matching alphanumeric groups of controlled length.
Measures: Rapid classification and comparison speed.

Module 3 – Timed Complexity Challenge
Students complete increasingly dense visual searches under strict time constraints.
Measures: Speed–accuracy trade-off and processing efficiency.

Module 4 – Adaptive Difficulty Engine
Stimulus density, distractor similarity, and presentation speed adapt to performance.
Measures: Processing efficiency across varying levels of complexity.

5. Recommended Scoring Model

• Perceptual Speed Score – 30%

• Visual Scanning Efficiency Score – 25%

• Rapid Classification Score – 20%

• Speed–Accuracy Trade-off Score – 25%

6. AI Analytics to Capture

• Average reaction time

• Response latency distribution

• Correct responses

• Commission errors

• Omission errors

• Fatigue slope

• Speed consistency index

• Accuracy trends

• Adaptive progression curve

• Processing efficiency profile

7. MVP Deliverables (Current Phase)

✓ Random sequence generator

✓ Minimum 5,000 unique sequence combinations

✓ Adaptive difficulty engine

✓ 120-second assessment timer

✓ Raw clickstream exporter

✓ Event logger

✓ Analytics dashboard

✓ Documentation

✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:
student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:
event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:
raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:
behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:
Refine the perceptual speed assessment to improve construct purity and develop the foundational MVP prototype.

Expected Deliverables:
1. Literature review on Processing Speed (Gs) and CHC theory.
2. Construct map defining perceptual speed, visual scanning, and rapid classification.
3. Design specifications for all assessment modules.
4. Random sequence generator prototype.
5. Adaptive difficulty specifications.
6. Clickstream data schema and event logging implementation.
7. Analytics dashboard design.
8. Demonstration prototype of one adaptive module.

10. Definition of Done (DoD)

Scientific:
✓ Processing Speed components clearly defined.
✓ Working Memory demands minimized.
✓ Stimulus complexity calibrated.

Technical:
✓ Random sequence generator implemented.
✓ Minimum 5,000 unique combinations generated.
✓ Adaptive difficulty engine functional.
✓ Event logging and analytics dashboard available.
✓ Results exported in SDK format.

Validation:
✓ Average assessment completion time below 2 minutes.
✓ Internal consistency ≥ 0.80.
✓ Pilot study with at least 100 students.
✓ Preliminary evidence of Processing Speed measurement.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Processing Speed that follows the common Assessment SDK. Focus on creating rapid visual discrimination tasks, adaptive complexity mechanisms, and rich behavioral analytics. This module will later be integrated into the unified cognitive and career assessment platform.